$$\begin{bmatrix} 1 & -4 & 7 & -5 & 0 \\ 0 & 1 & -4 & 3 & 0 \\ 2 & -6 & 6 & -4 & 0 \end{bmatrix} \xrightarrow[2\times Row1]{Row3=Row3-} \begin{bmatrix} 1 & -4 & 7 & -5 & 0 \\ 0 & 1 & -4 & 3 & 0 \\ 0 & 2 & -8 & 6 & 0 \end{bmatrix} \xrightarrow[-2\times Row2]{Row3=Row3}$$

$$\begin{bmatrix} 1 & -4 & 7 & -5 & 0 \\ 0 & 1 & -4 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow[+4\times Row2]{Row1=Row1} \begin{bmatrix} 1 & 0 & -9 & 7 & 0 \\ 0 & 1 & -4 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Verification of whether the final matrix is in rref

1. Row 1 and 2 are non-zero rows. Row 3 is a zero row. The condition that all the non-zero rows must be above the zero rows is True.
2. Row 1 Leading element = 1 in Column 1
   Row 2 Leading element = 1 in Column 2
   ∴ The condition that for each row, the leading entry must be in a column to the right of the columns of the leading elements of all rows above it is True.
3. Row 2 Column 1 = Row 3 Column 1 = 0
   Row 3 Column 2 = 0
   ∴ The condition that for each column having a leading element, all positions below it must contain 0 is True.
4. The condition that each leading element must be 1 is True.
5. Row 1 Column 2 = 0
   ∴ The condition that for each column having a leading element, all positions above it must contain 0 is True.

∴ The matrix $\begin{bmatrix} 1 & 0 & -9 & 7 & 0 \\ 0 & 1 & -4 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ is in rref. □

Column 3 and 4 have no pivots. By defn, ∴ x_3 and x_4 are free variables. (By defn.)

By Theorem 2, the system is consistent and has infinitely many solutions.

The corresponding system of equations are: ① $x_1 - 9x_3 + 7x_4 = 0$ ② $x_2 - 4x_3 + 3x_4 = 0$ ③ $0 = 0$

∴ $x_1 = 9x_3 - 7x_4$ and $x_2 = 4x_3 - 3x_4$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 9x_3 - 7x_4 \\ 4x_3 - 3x_4 \\ x_3 \\ x_4 \end{bmatrix} = x_3 \begin{bmatrix} 9 \\ 4 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -7 \\ -3 \\ 0 \\ 1 \end{bmatrix}$$ □